

# Hamza Muhammad Sohail

Senior Full Stack Engineer | Node.js | NestJS | React | Angular | TypeScript | AWS

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## PROFESSIONAL SUMMARY

Senior Full Stack Engineer with 5 years of experience building scalable web applications across IoT, healthcare, recruitment and e-learning. Strong in backend development with Node.js, NestJS, Express.js and TypeScript, designing REST APIs, asynchronous and event-driven services, and real-time data pipelines on AWS. Builds responsive front ends in React, Redux Toolkit and Angular that integrate cleanly with backend services. Experienced in client-facing requirement gathering, solution design, code reviews and mentoring. Uses AI-assisted, spec-driven development (Claude, GitHub Copilot, OpenSpec) to deliver features faster without sacrificing quality. BS Computer Science Gold Medalist (CGPA 3.9).

## TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript, Python, C#, SQL

**Backend:** Node.js, NestJS, Express.js, Koa.js, ASP.NET, REST APIs, JWT, Passport, Swagger / OpenAPI, Serverless Framework

**Frontend:** React, Redux Toolkit, Redux, Angular 17, Angular Material, AngularJS, Vite, Mapbox

**Databases:** PostgreSQL, PostGIS, MongoDB, MySQL, Redis, Elasticsearch, InfluxDB

**Cloud and DevOps:** AWS (IoT Core, Lambda, S3, SES, SQS, SNS, CloudWatch, ECR), Docker, CircleCI, Nx, Flyway, Git, GitHub, Bitbucket

**Real-Time and Integrations:** WebRTC (PexRTC), Twilio, PubNub, Pub/Sub, Stripe

**Testing and Monitoring:** Jest, Unit and Integration Testing, Grafana k6, SonarQube, ESLint, New Relic, Elastic Stack (ELK)

**AI-Assisted Development:** Claude, GitHub Copilot, OpenSpec, Spec-Driven Development

## PROFESSIONAL EXPERIENCE

### 10Pearls Pakistan

Full-time • 4 yrs 7 mos

#### Senior Software Consultant

Feb 2025 – Present • 1 yr 8 mos | Lahore, Pakistan • Hybrid

*Connected RV IoT Platform: multi-tenant fleet and device management (web and mobile portals)*

- Delivered a feature estimated at 160 hours in 77.5 hours (52% faster) by adopting spec-driven development with OpenSpec, Claude and GitHub Copilot, locking requirements into reviewable specs before implementation.
- Developed multi-tenant NestJS services for device onboarding, lifecycle state and subscription management across web and mobile portals.
- Ingested real-time GPS and sensor telemetry through AWS IoT Core; processed mileage and geospatial data with PostgreSQL/PostGIS and stored time-series readings in InfluxDB for operational dashboards and analytics.
- Built REST APIs secured with JWT and Passport and documented with Swagger; offloaded asynchronous work to AWS SQS/SNS queues and Serverless Framework Lambda functions.
- Delivered React and Redux Toolkit portal screens, integrating Mapbox for live fleet mapping and Stripe for subscription billing.
- Owned client-facing requirement gathering and solution design; maintained code quality with Jest, ESLint and SonarQube and managed database schema changes with Flyway migrations.

## Software Consultant

Mar 2023 – Feb 2025 · 2 yrs | Lahore, Pakistan

*Hiring Platform: recruitment single-page application for job posting, candidate search and applications*

- Developed and maintained Node.js REST APIs (Koa.js, Express.js) for job posting, application tracking and candidate search, using Elasticsearch for fast filtering across large candidate sets.
- Delivered Angular 17 and Angular Material features in TypeScript that improved responsiveness across candidate and employer journeys.
- Secured storage of CVs and attachments with AWS S3 bucket policies and used Redis for caching across PostgreSQL and MongoDB data stores.
- Ran load tests with Grafana k6 and monitored services with AWS CloudWatch and the Elastic Stack (Elasticsearch, Logstash, Kibana); shipped containerized services with Docker and AWS ECR through CI/CD pipelines.

## Associate Software Engineer

Mar 2022 – Mar 2023 · 1 yr 1 mo | Karachi, Pakistan

*Health Care Solution: telemedicine platform for remote doctor visits and virtual consultations*

- Built dynamic camera and microphone switching for live video consultations using PexRTC (WebRTC), with page display configurable per deployment.
- Refactored callback and promise chains to async/await, removing blocking operations and improving stability and response times under concurrent consultation load.
- Optimized long-running PostgreSQL queries for the reporting module and exposed REST APIs for other applications to consume platform data.
- Upgraded Node.js and New Relic versions to close security vulnerabilities, and migrated logging from Pino to ECS format for Elasticsearch compatibility.
- Wrote and optimized unit and integration tests, and delivered AngularJS front-end features to improve interactivity.
- Identified and resolved build failures across environments in CircleCI, and managed the application's search-engine visibility per environment.
- Met directly with clients to gather requirements, share progress updates and address concerns.
- Created, prioritized and assigned issues for tasks, bugs and user stories and tracked them through to done; mentored junior team members and supported teammates on Python tasks.

## Tafsol Technologies

### Node.js Developer

Sep 2021 – Jan 2022 · 5 mos | Karachi, Pakistan

*MentGo (mentorship platform) and Online Medical Learning Platform (subscription video learning)*

- Built Node.js and Express.js REST APIs with MongoDB for mentor, booking, catalogue, playback and progress data.
- Implemented queue-based email notifications, time zone conversion for mentor and mentee bookings, and commission calculations for paid sessions.
- Enforced subscription-based content access and one-device lifetime access at the API layer.
- Secured video content with AWS S3 bucket policies to prevent unauthorized access to media URIs; used Redis for caching.

## EDUCATION

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**Bachelor of Science in Computer Science | Bahria University, Karachi Campus | 2018 – 2022**

CGPA 3.9 / 4.0 | Gold Medalist